

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 12- Statistics

October, 2023.

1. A computer company received a rush order for as many home computers as could be shipped during a six week period. Company records provide the following daily shipments.

22	65	65	67	55	85	65
77	73	30	62	54	48	65
79	60	63	45	51	68	79
83	33	41	49	28	55	61
65	75	55	75	39	87	45
50	66	65	59	25	35	53

- (a) Group these daily shipment figures into a frequency distribution having the suitable number of classes.
- (b) Draw a frequency histogram to illustrate these data.
- (c) Draw a frequency polygon for this information.
2. The following are the number of two wheelers sold by a dealer during 8 weeks of 6 working days each.

13	19	22	14	13	16	19	21
23	11	27	25	17	17	13	20
23	17	26	20	24	15	20	21
23	17	29	17	19	14	20	20
10	22	18	25	16	23	19	20
21	17	18	24	21	20	19	26

- (a) Group these figures into a table having the classes 10 – 12, 13 – 15, 16 – 18, ...
- (b) Find the cumulative frequency distribution.
- (c) Draw a cumulative frequency curve for this information.
3. Two tutorials groups that wrote the same MAT1110 quiz 3 had the following results:

Group 9 : 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68
Group 10 : 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84

For each group, calculate

- (a) Range.

- (b) Mean.
- (c) Median.
- (d) Mode

4. The following table lists the daily temperatures of a certain town. Calculate the range of this data set.

-15 -6 -2 -8 -18 -21 -24 -25 -24 -25

5. Calculate the mean for the following frequency distribution.

Mass(g)	1 – 20	21 – 40	41 – 60	61 – 80	81 – 100
Number of letters	10	18	24	14	18

6. Develop a histogram for the data below, the mileage of a specific car with a full tank:

302 315 265 296 289 301 308 280 285 318
 267 300 309 312 299 316 301 286 281 311
 272 295 305 283 309 313 278 286 296 291
 310 302 282 287 307 305 314 318 318 280

7. An advertising company kept an account of response letters received each day over a period of 50 days. The observations were as follows:

0 2 1 1 1 2 0 0 1 0 1 0 0 1 0 1 1 0
 2 0 0 2 0 1 0 1 0 1 0 3 1 0 1 0 1 0
 2 5 1 2 0 0 0 0 5 0 1 1 2 0

Construct a frequency table and draw a histogram for this information.

8. Class A: 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68
 Class B: 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84
- (a) State the mode for each class.
 - (b) Calculate the median for each class.

9. A group of University students took part in a sponsored race. The number of laps completed is given in the table below. Use the information to:

Number of laps	Frequency (x)
1 – 5	2
6 – 10	9
11 – 15	15
16 – 20	20
21 – 25	17
26 – 30	25
31 – 35	2
36 – 40	1

- (a) Calculate an estimate for the mean number of laps.
- (b) Determine the modal class.

- (c) Determine the class interval containing the median.
- (d) Compute the mode and the median.
10. Two tutorials groups that wrote the same MAT1110 quiz 3 had the following results:
Group 5: 71, 82, 55, 76, 66, 71, 90, 84, 95, 64, 71, 70, 83, 45, 73, 51, 68
Group 6: 54, 80, 12, 61, 73, 69, 92, 81, 80, 61, 75, 74, 15, 44, 91, 63, 50, 84 Calculate the variance and standard deviation for each group.
11. Calculate the variance and standard deviation for the frequency distribution below.

Classes	Frequency
1 – 10	11
10 – 20	29
20 – 30	18
30 – 40	4
40 – 50	5
50 – 60	3